Assignment 8

;Write 8086/64 Assembly language program (ALP) to perform non-overlapped block
;transfer (With and without string specific instructions). Block Containing data can be
;defined in the data Segment.

```
%macro IO 4
        mov eax,%1
        mov ebx,%2
        mov ecx,%3
        mov edx,%4
        int 80h
%endmacro

;*****************************************************************
section .data
        menumsg db 10,'---Menu for Non-overlapped Block Transfer---',10
                db 10,'1.Block Transfer without using string instructions'
                db 10,'2.Block Transfer with using string instructions'
                db 10,'3.Exit',10
        menumsg_len equ $-menumsg

        blk_bfrmsg db 10,'Block contents before transfer'
        blk_bfrmsg_len equ $-blk_bfrmsg

        blk_afrmsg db 10,'Block contents after transfer'
        blk_afrmsg_len equ $-blk_afrmsg

        srcmsg db 10,'Source block contents::'
        srcmsg_len equ $-srcmsg

        dstmsg db 10,'Destination block contents::'
        dstmsg_len equ $-dstmsg

        srcblk db 01h,02h,03h,04h,05h
```

```
        dstblk db 00,00,00,00,00

        space db " "

;****************.bss Section**************************
section .bss

        choice resb 02
        output resb 02
        count resb 02

;*******************.text Section**************************

section .text
        global _start
_start:

        IO 4,1,blk_bfrmsg,blk_bfrmsg_len
        call display
menu:
        IO 4,1, menumsg,menumsg_len
        IO 3,0, choice,02
        cmp byte[choice],'1'
        je wos
        cmp byte[choice],'2'
        je ws

exit:
        mov eax,1       ;Exit
        mov ebx,0
        int 80h
```

```
;**************Display Content Procedure****************
disp_content:
        mov byte [count],05h
next:
        mov bl,[ebp]
        call disp8_proc
        IO 4,1, space,1
        inc ebp
        dec byte [count]
        jnz next
        ret

;**************Display Block Procedure*************
display:
        IO 4,1, srcmsg,srcmsg_len
        mov ebp,srcblk
        call disp_content
        IO 4,1, dstmsg,dstmsg_len
        mov ebp,dstblk
        call disp_content
        ret

;**************Without String Procedure************
wos:
        mov esi,srcblk
        mov edi,dstblk
        mov ecx,05

again:
        mov bl,[esi]
        mov [edi],bl
        inc esi
```

```
        inc edi
        loop again
        IO 4,1, blk_afrmsg,blk_afrmsg_len
        call display
        jmp menu

;************Using String Procedure****************
ws:
        mov esi,srcblk
        mov edi,dstblk
        mov ecx,05

        cld
        rep movsb
        IO 4,1, blk_afrmsg,blk_afrmsg_len
  call display
        jmp menu

;************Display Output Procedure*******************
disp8_proc:
        mov esi,output
        mov ecx,02
up1:
        rol bl,4
        mov dl,bl
        and dl,0Fh
        cmp dl,09H
        jbe skip
        add dl,07h

skip:
        add dl,30h
```

```
mov [esi],dl
inc esi
loop up1

IO 4,1, output,02
ret
```

//OUTPUT

```
pllab0112@pllab0112-ThinkCentre-M70s: ~/Documents/NMIE...
pllab0112@pllab0112-ThinkCentre-M70s:~/Documents/NMIET/7$ nasm -f elf nonoverlap
.asm
pllab0112@pllab0112-ThinkCentre-M70s:~/Documents/NMIET/7$ ld -m elf_i386 -o nono
verlap  nonoverlap.o
pllab0112@pllab0112-ThinkCentre-M70s:~/Documents/NMIET/7$ ./nonoverlap

Block contents before transfer
Source block contents::01 02 03 04 05
Destination block contents::00 00 00 00 00
---Menu for Non-overlapped Block Transfer---

1.Block Transfer without using string instructions
2.Block Transfer with using string instructions
3.Exit
1

Block contents after transfer
Source block contents::01 02 03 04 05
Destination block contents::01 02 03 04 05
---Menu for Non-overlapped Block Transfer---

1.Block Transfer without using string instructions
2.Block Transfer with using string instructions
3.Exit
2

Block contents after transfer
Source block contents::01 02 03 04 05
Destination block contents::01 02 03 04 05
---Menu for Non-overlapped Block Transfer---

1.Block Transfer without using string instructions
2.Block Transfer with using string instructions
3.Exit
3
pllab0112@pllab0112-ThinkCentre-M70s:~/Documents/NMIET/7$
```